

RAILGUARD

AI-Based Railway Track Risk & Maintenance Prioritization System

1. Problem Statement

Railway networks contain a large number of track sections that require continuous inspection and maintenance. Existing inspection systems can detect and monitor track defects, but authorities also need to decide which problematic section requires attention first, how urgent it is, and how the risk may change if maintenance is delayed.

2. Proposed Solution

RailGuard is a software-based decision-support system that combines track-condition data with factors such as defect severity, train traffic, weather, track age and maintenance history. It calculates a risk score, ranks track sections by priority, predicts short-term risk trends and provides explainable maintenance alerts.

3. What Makes RailGuard Different?

Existing systems mainly focus on detecting and monitoring track defects. RailGuard focuses on the decision-support layer after condition information is available. It combines multiple factors, ranks problematic sections, predicts 7-day/14-day risk trends, considers recurring defects and previous repairs, and explains why a section is high-risk.

4. Main Inputs / Data

- Track ID and location
- Defect type and severity
- Train frequency / traffic level
- Track age
- Weather and rainfall information
- Previous maintenance and repair history
- Number of recurring defects
- Historical condition/risk records

5. Core Features

- Risk Score: Generates a 0–100 risk score.

- Risk Classification: Low, Medium, High, or Critical.
- Priority Ranking: Identifies which track needs attention first.
- 7-Day / 14-Day Risk Prediction.
- Maintenance History and recurring-defect detection.
- Explainable Alerts showing the main reasons for high risk.
- Maintenance Priority Recommendation.
- Map-Based Dashboard with color-coded risk levels.

6. Example

Track T001 has a medium defect, high train traffic, repeated previous repairs and recent heavy rainfall.

Example output:

Current Risk: 72/100

7-Day Predicted Risk: 84/100

14-Day Predicted Risk: 91/100

Priority: #1

Reasons: High traffic, recurring defect, rainfall and previous repairs

Recommendation: High-priority inspection/maintenance.

7. How We Will Build It

Step 1 — Dataset: Prepare structured track-condition, traffic, weather and maintenance data. Simulated/curated data can be used for the prototype if real operational data is unavailable.

Step 2 — Data Processing: Use Python and Pandas to clean and prepare the data.

Step 3 — Risk Engine: Build a transparent weighted risk-scoring model using factors such as defect severity, traffic, maintenance history, weather and track age.

Step 4 — Priority Engine: Rank track sections according to risk and impact.

Step 5 — Machine Learning: Add a model such as Random Forest/XGBoost for risk prediction after the baseline works.

Step 6 — Forecasting: Use historical condition/risk records to estimate 7-day and 14-day risk trends.

Step 7 — Backend: Use FastAPI or Flask to connect the models with the application.

Step 8 — Dashboard: Build a React dashboard showing risk, priority, history, alerts and explanations.

Step 9 — Map: Display track sections using Green, Yellow, Orange and Red risk markers.

Step 10 — Integration: Connect all modules and test different track scenarios.

8. Suggested Technology Stack

Python + Pandas — Data processing

Scikit-learn / XGBoost — Machine Learning

FastAPI / Flask — Backend

React.js — Dashboard

Firebase / PostgreSQL / MySQL — Database

Google Maps API / OpenStreetMap — Map

GitHub — Version control

9. Software or Hardware?

The core SIH prototype will be software-based. Hardware is not compulsory. In a future version, RailGuard can receive data from cameras, IoT sensors, GPS or train-mounted inspection equipment. This makes the system compatible with real railway data sources without making hardware necessary for the first prototype.

10. System Architecture

Data Sources → Data Processing → Risk Engine → ML Prediction → Priority Engine → Backend/API → Dashboard + Map → Maintenance Alert

11. Prototype Scope

The first working version should demonstrate five things: track data input, risk calculation, priority ranking, maintenance-history display and dashboard output. Advanced ML prediction, maps and additional features can be added incrementally.

12. Expected Outcome

RailGuard will provide maintenance teams with a single view of track risk, priority, maintenance history, risk trends and recommended attention. It is designed to support earlier and better-informed maintenance decisions, not to replace official railway inspection or engineering judgment.

13. Novelty / USP

Existing railway technologies already perform track inspection and defect detection. RailGuard's proposed differentiation is the decision-support layer: multi-factor risk assessment, priority ranking, short-term risk forecasting, maintenance-history analysis and explainable recommendations.

14. One-Line USP

“Existing systems focus on detecting railway track problems; RailGuard focuses on prioritizing those problems and predicting how their risk may evolve if maintenance is delayed.”